

Jaron Arbet, Ph.D.

ASSISTANT RESEARCH PROFESSOR - BIOSTATISTICIAN

City of Hope - Beckman Research Institute - Department of Population Sciences

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Research Interests

- Machine learning and predictive modeling
- High-dimensional “Big Data”
- Variable selection
- Multi-omics cancer data integration
- Robust nonparametric statistics

Work Experience

Assistant Research Professor - Biostatistician

Dec 2025 - Present

CITY OF HOPE - BECKMAN RESEARCH INSTITUTE - DEPARTMENT OF POPULATION SCIENCES, DUARTE, CA

- Studying the impact of exercise on risk of cancer diagnosis, and for improving health outcomes in cancer patients.
- Analysis of cancer molecular data: whole genome sequencing, RNA-Seq, DNA methylation, proteomics.

Senior Bioinformatician

April 2022 - Nov 2025

JONSSON COMPREHENSIVE CANCER CENTER, UNIVERSITY OF CALIFORNIA LOS ANGELES, CA

- Analysis of cancer molecular data: whole genome sequencing, RNA-Seq, DNA methylation, proteomics
- Biomarker identification: among thousands of molecular features, which are most associated with cancer outcomes?
- Daily git/github use for version control and collaboration
- Supervised machine learning for predicting cancer prognosis
- Unsupervised machine learning for deriving cancer subtypes
- Supervising undergraduate and graduate students on projects
- Develop open-source R packages for sharing software with broader scientific community
- Create statistical analysis plans and power/sample size estimation for grants

Staff Biostatistician

Jan 2021 - April 2022

EDWARDS LIFESCIENCES, IRVINE, CA

- Develop statistical analysis plans and data monitoring for clinical trials
- Power and sample size estimation
- Clinical study reports for federal regulatory agencies
- Perform analyses with large national database (STS/ACC TVT Registry) including propensity score matching for causal inference

Postdoctoral Research Associate

Aug 2018 - Jan 2021

UNIVERSITY OF COLORADO ANSCHUTZ MEDICAL CAMPUS, DEPARTMENT OF BIOSTATISTICS AND INFORMATICS, AURORA, CO

- Provided statistical consulting services
- Published scientific manuscripts
- Developed grants (statistical analysis plan, power/sample size estimation)
- Supervised students and MS/MPH degree biostatisticians
- Taught BIOS 6601 Intro stats course for 75 MPH students, and created annual 6-week data science machine learning short course
- Develop new statistical methodology

Research Assistant

June 2014 - July 2018

UNIVERSITY OF MINNESOTA, DIVISION OF BIOSTATISTICS, MINNEAPOLIS, MN

- Developed variable selection methods for high dimensional genetic data and robust methods for estimating the genetic heritability of disease traits

Statistical Genetics Research Intern

June - Aug 2013

DORDT COLLEGE, DEPARTMENT OF MATHEMATICS, SIOUX CENTER, IA

- Developed methods for aggregating different statistical tests together for greater sensitivity in detecting disease-related genes; resampling-based bias corrected tests

Math and Statistics Tutor

Sept 2012 - May 2014

WINONA STATE UNIVERSITY, WINONA, MN

Manscripts submitted for review

1. **Arbet, J.** et al. (2025). The landscape of prostate tumour methylation. *bioRxiv*. <https://doi.org/10.1101/2025.02.07.637178>
2. Haas, R. et al. (2025). Diverse genomes, shared health: Insights from a health system biobank. *medRxiv*. <https://doi.org/10.1101/2025.06.11.25329386>

Peer-reviewed Publications

1. Jones, L. W., Lavery, J. A., Tsai, B. L., ... Boutros, P. C. (2025). A co-clinical trial of exercise therapy in breast cancer prevention. *Clinical Cancer Research*, 31(16), 3377–3387. <https://doi.org/10.1158/1078-0432.ccr-24-4298>
2. Xu, X., Zhu, H., Hugh-White, R., ... He, H. H. (2025). The landscape of N6-methyladenosine in localized primary prostate cancer. *Nature Genetics*, 57(4), 934–948. <https://doi.org/10.1038/s41588-025-02128-y>
3. Tohi, Y., Sahrman, J. M., **Arbet, J.**, ... Boutros, P. C. (2025). De-escalation of monitoring in active surveillance for prostate cancer: Results from the GAP3 consortium. *European Urology Oncology*, 8(2), 347–354. <https://doi.org/10.1016/j.euo.2024.07.006>
4. Lippitt, W., Carlson, N. E., **Arbet, J.**, ... Kechris, K. (2024). Limitations of clustering with PCA and correlated noise. *Journal of Statistical Computation and Simulation*, 94(10), 2291–2319. <https://doi.org/10.1080/00949655.2024.2329976>
5. Weiner, A. B., Agrawal, R., Wang, N. K., ... Reiter, R. E. (2024). Molecular hallmarks of prostate-specific membrane antigen in treatment-naïve prostate cancer. *European Urology*, 86(6), 579–587. <https://doi.org/10.1016/j.eururo.2024.09.005>
6. Ni, K., Rogowitz, E., Farahmand, A. K., ... Saxon, D. R. (2024). Weight loss outcomes in a veterans affairs pharmacotherapy-based weight management clinic. *Journal of the Endocrine Society*, 8(5). <https://doi.org/10.1210/jendso/bvae042>
7. Greca, A. L., Grau, L., **Arbet, J.**, ... Kitahara, C. M. (2023). Anthropometric, dietary, and lifestyle factors and risk of advanced thyroid cancer: The NIH-AARP diet and health cohort study. *Clinical Endocrinology*, 99(6), 586–597. <https://doi.org/10.1111/cen.14970>
8. Gamallat, Y., Choudhry, M., Li, Q., ... Bismar, T. A. (2023). Serrate RNA effector molecule (SRRT) is associated with prostate cancer progression and is a predictor of poor prognosis in lethal prostate cancer. *Cancers*, 15(10), 2867. <https://doi.org/10.3390/cancers15102867>
9. Chan, T. W., Dodson, J. P., **Arbet, J.**, ... Xiao, X. (2022). Single-cell analysis in lung adenocarcinoma implicates RNA editing in cancer innate immunity and patient prognosis. *Cancer Research*, 83(3), 374–385. <https://doi.org/10.1158/0008-5472.can-22-1062>
10. Okamoto, Y., Devoe, S., Seto, N., ... Demoruelle, M. K. (2021). Association of sputum neutrophil extracellular trap subsets with IgA anti-citrullinated protein antibodies in subjects at risk for rheumatoid arthritis. *Arthritis & Rheumatology*, 74(1), 38–48. <https://doi.org/10.1002/art.41948>
11. Lin, N. W., **Arbet, J.**, Mroz, M. M., ... Maier, L. A. (2022). Clinical phenotyping in sarcoidosis using cluster analysis. *Respiratory Research*, 23(1). <https://doi.org/10.1186/s12931-022-01993-z>
12. Grau, L., **Arbet, J.**, Ostendorf, D. M., ... Creasy, S. A. (2022). Creating an algorithm to identify indices of sleep quantity and quality from a wearable armband in adults. *Sleep Science*, 15(03), 279–287. <https://doi.org/10.5935/1984-0063.20220052>
13. Creasy, S. A., Ostendorf, D. M., Blankenship, J. M., ... Catenacci, V. A. (2022). Effect of sleep on weight loss and adherence to diet and physical activity recommendations during an 18-month behavioral weight loss intervention. *International Journal of Obesity*, 46(8), 1510–1517. <https://doi.org/10.1038/s41366-022-01141-z>
14. Wood, C., **Arbet, J.**, Amura, C. R., ... Anderson, J. (2021). Multicenter study evaluating nitrous oxide use for labor analgesia at high- and low-altitude institutions. *Anesthesia & Analgesia*, 134(2), 294–302. <https://doi.org/>

15. Rosenthal, L., Lee, S., Jenkins, P., ... Nodine, P. (2021). A survey of mental health in graduate nursing students during the COVID-19 pandemic. *Nurse Educator*, 46(4), 215–220. <https://doi.org/10.1097/nne.0000000000001013>
16. Reed, S. M., **Arbet, J.**, & Staubli, L. (2021). Clinical nurse specialists in the united states registered with a national provider identifier. *Clinical Nurse Specialist*, 35(3), 119–128. <https://doi.org/10.1097/nur.0000000000000592>
17. **Arbet, J.**, Zhuang, Y., Litkowski, E., ... Kechris, K. (2021). Comparing statistical tests for differential network analysis of gene modules. *Frontiers in Genetics*, 12. <https://doi.org/10.3389/fgene.2021.630215>
18. Nodine, P. M., **Arbet, J.**, Jenkins, P. A., ... Hoon, S. (2021). Graduate nursing student stressors during the COVID-19 pandemic. *Journal of Professional Nursing*, 37(4), 721–728. <https://doi.org/10.1016/j.profnurs.2021.04.008>
19. Thomas, E. A., Zaman, A., Cornier, M.-A., ... Rynders, C. A. (2020). Later meal and sleep timing predicts higher percent body fat. *Nutrients*, 13(1), 73. <https://doi.org/10.3390/nu13010073>
20. **Arbet, J.**, Brokamp, C., Meinen-Derr, J., ... Spratt, H. M. (2020). Lessons and tips for designing a machine learning study using EHR data. *Journal of Clinical and Translational Science*, 5(1). <https://doi.org/10.1017/cts.2020.513>
21. Ramakrishnan, V. R., **Arbet, J.**, Mace, J. C., ... Smith, T. L. (2021). Predicting olfactory loss in chronic rhinosinusitis using machine learning. *Chemical Senses*, 46. <https://doi.org/10.1093/chemse/bjab042>
22. Ostendorf, D. M., Blankenship, J. M., Grau, L., ... Catenacci, V. A. (2021). Predictors of long-term weight loss trajectories during a behavioral weight loss intervention: An exploratory analysis. *Obesity Science & Practice*, 7(5), 569–582. <https://doi.org/10.1002/osp4.530>
23. Schmanski, A., Roberts, E., Coors, M., ... Taylor, M. R. G. (2020). Research participant understanding and engagement in an institutional, self-consent biobank model. *Journal of Genetic Counseling*, 30(1), 257–267. <https://doi.org/10.1002/jgc4.1316>
24. Carpenter, C. M., Frank, D. N., Williamson, K., ... Kroehl, M. E. (2021). tidyMicro: A pipeline for microbiome data analysis and visualization using the tidyverse in r. *BMC Bioinformatics*, 22(1). <https://doi.org/10.1186/s12859-021-03967-2>
25. **Arbet, J.**, McGue, M., & Basu, S. (2020). A robust and unified framework for estimating heritability in twin studies using generalized estimating equations. *Statistics in Medicine*, 39(27), 3897–3913. <https://doi.org/10.1002/sim.8564>
26. Coleman-Minahan, K., Sheeder, J., **Arbet, J.**, (2020). Interest in medication and aspiration abortion training among colorado nurse practitioners, nurse midwives, and physician assistants. *Women's Health Issues*, 30(3), 167–175. <https://doi.org/10.1016/j.whi.2020.02.001>
27. Gance-Cleveland, B., Linton, A., **Arbet, J.**, ... Sylvain, G. (2020). Predictors of overweight and obesity in childhood cancer survivors. *Journal of Pediatric Oncology Nursing*, 37(3), 154–162. <https://doi.org/10.1177/1043454219897102>
28. James-Allan, L. B., **Arbet, J.**, Teal, S. B., ... Jansson, T. (2019). Insulin stimulates GLUT4 trafficking to the syncytiotrophoblast basal plasma membrane in the human placenta. *The Journal of Clinical Endocrinology & Metabolism*, 104(9), 4225–4238. <https://doi.org/10.1210/jc.2018-02778>
29. Grinde, K. E., **Arbet, J.**, Green, A., ... Tintle, N. (2017). Illustrating, quantifying, and correcting for bias in post-hoc analysis of gene-based rare variant tests of association. *Frontiers in Genetics*, 8. <https://doi.org/10.3389/fgene.2017.00117>
30. **Arbet, J.**, McGue, M., Chatterjee, S., (2017). Resampling-based tests for lasso in genome-wide association studies. *BMC Genetics*, 18(1). <https://doi.org/10.1186/s12863-017-0533-3>
31. Greco, B., Hainline, A., **Arbet, J.**, ... Tintle, N. (2015). A general approach for combining diverse rare variant association tests provides improved robustness across a wider range of genetic architectures. *European Journal of Human Genetics*, 24(5), 767–773. <https://doi.org/10.1038/ejhg.2015.194>

Awards

POSTDOCTORAL

Staff appreciation and recognition (STAR)

UCLA JONSSON COMPREHENSIVE CANCER CENTER

2023-2024

Staff appreciation and recognition (STAR)

UCLA JONSSON COMPREHENSIVE CANCER CENTER

2022-2023

STUDENT AWARDS

Outstanding Research Assistant Award

UNIVERSITY OF MINNESOTA, DIVISION OF BIOSTATISTICS & HEALTH DATA SCIENCE

2017-2018

Interdisciplinary Biostatistics Training Grant in Genetics and Genomics

NATIONAL INSTITUTE OF GENERAL MEDICAL SCIENCES

2015-2017

COMPETITIONS

Midwest Undergraduate Data Analysis Competition (MUDAC)

SCHOOL OF EDUCATION, UNIVERSITY OF MICHIGAN

- 3rd place out of 20 teams from 14 universities

2014

1st place in the “predictive-modeling class competition”

WINONA STATE UNIVERSITY DATA MINING COURSE

2014

Midwest Undergraduate Data Analysis Competition (MUDAC)

SCHOOL OF EDUCATION, UNIVERSITY OF MICHIGAN

- 2nd place out of 15 teams from 10 universities

2013

Presentations

INVITED GUEST SPEAKER

The Landscape of Prostate Tumour Methylation

PROSTATE CANCER FOUNDATION PRECISION MEDICINE WORKING GROUP, VIRTUAL MEETING

- <https://www.biorxiv.org/content/10.1101/2025.02.07.637178v3>

2025

DEPARTMENTAL

Mendelian Randomization Causal inference in Observational Studies

STATISTICS WORKING GROUP, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- <https://jarbet.github.io/presentation-mendelian-randomization/>

2025

Batch effects: problem and solution

BOUTROS LAB, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentations/2025-05-08_batch_effects_nowhite.pdf

2025

p-value histograms

BOUTROS LAB, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- <https://jarbet.github.io/presentation-pvalue-histograms/>

2025

Robust regression for noisy biological data

STATISTICS WORKING GROUP, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentation-ordinal-regression/ordinal_regression.html

2024

Linear regression diagnostics

BOUTROS LAB, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentation-regression-diagnostics/regression_diagnostics.html

2024

Causal inference with observational data using propensity score matching

STATISTICS WORKING GROUP, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentation-causal-inference-matching/causal_inference_matching.html

2024

Automatic feature selection and engineering with MARS

STATISTICS WORKING GROUP, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- <https://jarbet.github.io/presentation-mars.github.io/presentation-mars.html>

2023

Introduction to Bayesian statistics

BOUTROS LAB, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentations/2022-08-25_Bayesian_stats.pdf

2022

Multiomics cancer data analysis

BOUTROS LAB, CANCER DATA SCIENCES, UNIVERSITY OF CALIFORNIA, LOS ANGELES, CA

- https://jarbet.github.io/presentations/2021-11-21_multiomics_cancer.pdf

2021

Interpretable machine learning

CENTER FOR INNOVATIVE DESIGN AND ANALYSIS, UNIVERSITY OF COLORADO, DENVER, CO

2021

- https://jarbet.github.io/presentations/2020-01-21_interpretable_machine_learning.pdf

Differential Network Analysis of Gene Modules

SYSTEMS GENETICS AND BIOINFORMATICS RESEARCH LAB, UNIVERSITY OF COLORADO, DENVER, CO

2021

Frequentist versus Bayesian Approaches to Statistics: A Debate

DEPARTMENT OF BIostatISTICS AND INFORMATICS, UNIVERSITY OF COLORADO, DENVER, CO

2018

Robust Estimation of Genetic and Environmental Variance Components in Twin Studies

UNIVERSITY OF MINNESOTA BIostatISTICS STUDENT SEMINAR SERIES, MINNEAPOLIS, MN

2017

A General framework for Association Tests with Multivariate Traits in Large-Scale Genomics Studies

UNIVERSITY OF MINNESOTA STATISTICAL GENETICS JOURNAL CLUB, MINNEAPOLIS, MN

2016

Methods of Inference for Penalized Regression in High-Dimensional Genetic Association Studies

SCHOOL OF PUBLIC HEALTH RESEARCH DAY, UNIVERSITY OF MINNESOTA, MINNEAPOLIS, MN

2016

Penalized Regression for High-Dimensional Genetic Association Testing

UNIVERSITY OF MINNESOTA STATISTICAL GENETICS JOURNAL CLUB, MINNEAPOLIS, MN

2016

■ CONFERENCES

The Landscape of Prostate Tumour Methylation

UCLA HEALTH JONSSON COMPREHENSIVE CANCER CENTER ANNUAL SYMPOSIUM, LOS ANGELES, CA

Poster

2025

Methylome-Genome Interactions Define Disease Aggression in Localized Prostate Cancer

RECOMB/ISCB CONFERENCE ON REGULATORY AND SYSTEMS GENOMICS, LOS ANGELES, CA

Poster

2024

Triplet matching: propensity score matching with 3 groups

14TH ANNUAL FDA/ADVA MEDICAL DEVICE STATISTICAL ISSUES CONFERENCE (VIRTUAL)

Talk

2022

- https://jarbet.github.io/presentations/20220422_triplet_matching.pdf

Statistical Considerations and Methods to Utilize Real World Evidence in Medical Device Evaluation

14TH ANNUAL FDA/ADVA MEDICAL DEVICE STATISTICAL ISSUES CONFERENCE (VIRTUAL)

Panel Co-organizer

2022

Robust Estimation of Genetic and Environmental Variance Components in Twin Studies using Generalized Estimating Equations

JOINT STATISTICAL MEETINGS (JSM), BALTIMORE, MD

Poster

2017

Methods of Inference for Penalized Regression in High-Dimensional Genetic Association Studies

EASTERN NORTH AMERICAN REGION INTERNATIONAL BIOMETRIC SOCIETY (ENAR), AUSTIN, TX

Poster

2016

What now? Post-Hoc Approaches for Gene-based Rare-Variant Tests of Association

AMERICAN SOCIETY OF HUMAN GENETICS CONFERENCE (ASHG), BOSTON, MA

Poster

2013

General Approaches for Combining Rare Variant Association Tests Provide Improved Power Across a Wide Range of Genetic Architectures

AMERICAN SOCIETY OF HUMAN GENETICS CONFERENCE (ASHG), BOSTON, MA

Poster

2013

Teaching Experience

■ SEMESTER COURSES

BIOS 6601 – Applied Biostatistics I

UNIVERSITY OF COLORADO ANSCHUTZ MEDICAL CAMPUS

Fall 2019

- **Duty:** Sole Instructor, completely redesigned all lectures, homework, quizzes, and exams
- **Description:** Applied biostatistical methods including descriptive and statistical inference; odds ratio and relative risk, probability theory, parameter estimation, tests for comparing statistics of two or more groups, correlation, linear regression, logistic regression and survival analysis.
- **Credits:** 3
- **Audience:** 75 Masters of Public Health (MPH) students

■ SHORT COURSES

Fundamentals of Data Science Literacy

Summer 2019, Fall 2020

UNIVERSITY OF COLORADO ANSCHUTZ MEDICAL CAMPUS

- **Duty:** lead organizer, developed and taught 3 machine learning lectures
- **Description:** 6-week short course (2 hours/week) on data science with emphasis on how machine learning and data mining are used in biomedical research. Topics included: 1) Overview of Data Science; 2) Introduction of Machine Learning in Biomedical Research; 3) Clustering and Pattern Finding (unsupervised Machine Learning); 4) Powerful Predictive Models and Variable Selection (supervised Machine Learning); 5) Fundamentals of Data Visualization; 6) Reproducible Research and Team Science
- **Audience:** 20 medical researchers, mostly non-statisticians

Introduction to Statistics

Summer 2019, 2020

UNIVERSITY OF COLORADO ANSCHUTZ MEDICAL CAMPUS

- **Duty:** Instructor, developed and taught lectures on “Introduction to statistics”, study design, and survey design
- **Description:** Introductory statistics day-long short course for UCHHealth nurses
- **Audience:** 15 UCHHealth Nurses

■ GUEST LECTURES

IPHY 5900 – Data Literacy in Biomedical Research

Nov 2019, Oct 2020

UNIVERSITY OF COLORADO BOULDER

- **Primary Instructor:** Dr. Celine Marie Claire D. Vetter, Integrative Physiology
- **Lecture:** Machine Learning in Biomedical Research

BIOS 6611 – Biostatistical Methods I

Fall 2018

UNIVERSITY OF COLORADO SCHOOL OF PUBLIC HEALTH

- **Primary Instructor:** Dr. Alex Kaizer
- **Lecture:** Linear Algebra Review

Mentoring and supervising

UNIVERSITY OF CALIFORNIA LOS ANGELES

Supervising students for data analysis projects

- Yuxi Song
- Sattwik Banerjee
- Adriana Wiggins

UNIVERSITY OF COLORADO, DEPARTMENT OF BIostatISTICS AND INFORMATICS

Supervising MPH/MS Statisticians

- Laura Grau (MPH), Nov 2018 – Jan 2021
 - I supervised Laura on several projects which led to publications.

Supervising students for data analysis projects

- Yaxu Zhuang (PhD), Jan 2019 – Aug 2020
- Rachel Weber (MS), Dec 2018 – April 2019

Thesis or Dissertation Committee Member (graduating year)

- Charlie Carpenter, Biostatistics MS thesis committee, Biostatistics (2020)
- Rachel Blumhagen, Biostatistics PhD dissertation committee (2021)

Professional Service

GRANT AND PROTOCOL REVIEWS

- Scientific Advisory & Review Committee (SARC), Colorado Clinical and Translational Sciences Institute, University of Colorado – April 2019 to Jan 2021

JOURNAL REVIEWER

- Biometrics
- Genetic Epidemiology
- PeerJ

COMMITTEES/OTHER

Jonsson Comprehensive Cancer Center, University of California Los Angeles

- Chair of Statistics Working Group: April 2023 - Present

Department of Biostatistics and Informatics, University of Colorado

- Student Admissions Committee, Fall 2018 – Spring 2019
- Volunteer Judge for CIDA Hackathon (24 hour data analysis competition), Jan. 2020

Biostatistics, Epidemiology, & Research Design Core of the Colorado Clinical and Translational Sciences Institute, University of Colorado

- Education Committee, Fall 2018 – Jan 2020

Software Skills

■ PROFICIENT



- Over **15 years** of experience with R for data analysis
- I enjoy **developing R packages** to easily share documented code with colleagues and the broader research community. Examples:
 - <https://uclahs-cds.github.io/package-PrCaMethy/>
 - <https://github.com/arbet003/discoMod>
- **Parallel programming** in R using multiple cpus

git/github

- Version control
- Used daily
- Collaborating with others on projects
- Review colleagues' code
- Assign issues to team members

Quarto and Rmarkdown

- Reproducible research, manuscripts and reports
- Reproducible HTML presentations. Examples: <https://jarbet.github.io/presentations>
- Websites: <https://jarbet.github.io/>
- Resume: this resume was made using Rmarkdown with the vitae R package

Bayesian statistical programming

- Proficient with BUGS/JAGS; basic skills with STAN

■ BASIC SKILLS

- Linux command line and bash scripts
 - Daily use of a high-performance computing cluster at UCLA
- Databricks
- Microsoft Office
- SAS